Quantification: 876_z82c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	8
interior lacuna count	6
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 6)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1771.5	1867	619.765	px^2
major_axis_length_px	lacuna shape	75.501	77.7477	19.5694	px
minor_axis_length_px	lacuna shape	30.3022	30.7562	5.812	px
aspect_ratio	lacuna shape	2.5004	2.4623	0.504	unitless
eccentricity	lacuna shape	0.9052	0.9137	0.0454	unitless
solidity	lacuna shape	0.8535	0.8701	0.0443	unitless
roots_count	headline	9.5	9.5	3.8341	count
ring_length_r30_px	headline	614.667	634	129.821	px
ring_length_r60_px	reported	1661.5	1819	391.825	px
owned_length_px	ownership-dependent	7006.18	7475.28	3290.47	px
edge_count	ownership-dependent	511.667	552	234.827	count
mean_edge_length_px	ownership-dependent	13.5903	13.5593	0.5779	px
perimeter_px	normalised	233.197	239.049	60.4736	px
ring_area_r30_px2	normalised	8429.67	8550	1388.13	px^2
ring_area_r60_px2	normalised	21961.5	22639.5	3572.96	px^2
in_frame_fraction_r30	normalised	0.9983	1	0.0042	unitless
in_frame_fraction_r60	normalised	0.9757	1	0.0596	unitless
ring_density_r30	normalised	0.0729429	0.0749053	0.0100687	px^-1
ring_density_r60	normalised	0.0755709	0.0805309	0.0122517	px^-1
roots_per_100px_perimeter	normalised	4.0214	4.3987	0.9994	per 100 px
ring_attached_length_r30_px	network v2	575.167	596.5	114.601	px
ring_attached_length_r60_px	network v2	1462.17	1645.5	414.198	px
ring_length_w_r30_px	network v2	694.18	721.786	143.098	px
ring_length_w_r60_px	network v2	1874.02	2064.33	443.386	px
sholl_crossings_r10	network v2	14.5	14.5	3.3912	count
sholl_crossings_r20	network v2	18.5	19.5	4.7645	count
sholl_crossings_r30	network v2	20.8333	20.5	3.3116	count
loop_count	loops, report only	82	95	49.4449	count
loop_edge_share	loops, report only	0.5153	0.5791	0.1463	share
ring_width_mean_r30_px	width	6.5913	6.7881	0.7575	px

Per field
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measure	value	units
analysed_area_px2	1.0316e+06	px^2
total_skeleton_length_px	68259	px
canalicular_length_density_per_px	0.0661683	px^-1
junction_count	2410	count
junction_density_per_px2	0.00233619	px^-2
lacuna_count	8	count

lacunae_per_px2	7.75e-06	px^-2
skeleton_component_count	645	count
mean_component_length_px	105.828	px
median_component_length_px	12	px
field_length_density_w_per_px	0.074322	px^-1
field_density_without_flagged_per_px	0.0656571	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	147	px
unattached_component_count	9	count
canalicular_length_density_attached_per_px	0.0661683	px^-1
field_length_density_with_unattached_per_px	0.0663108	px^-1
loop_count	725	count
loop_edge_share	0.4738	share
edge_count	5274	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.1769	px
width_median_px	6	px
width_p10_px	2.8284	px
width_p90_px	10	px
width_pixels_used	55083	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	4013	213.469	32.7614	6.5158	0.9882	0.5369	1.527	16.3571	878.108	True
2	2473	101.199	32.7068	3.0941	0.9463	0.7952	1.5403	149.487	378.777	False
3	2169	83.3807	34.4381	2.4212	0.9107	0.8757	-1.5585	303.736	321.658	False
4	1583	72.1147	28.8056	2.5035	0.9168	0.876	-1.5035	531.782	196.779	False
5	1826	64.8209	36.821	1.7604	0.823	0.9062	-1.5669	541.512	442.987	False
6	1908	86.6787	28.6695	3.0234	0.9437	0.8645	-1.4722	691.032	607.199	False
7	670	44.8117	20.3719	2.1997	0.8907	0.8034	-1.4095	908.587	49.1806	False
8								188.435	951.62	True

Network per lacuna

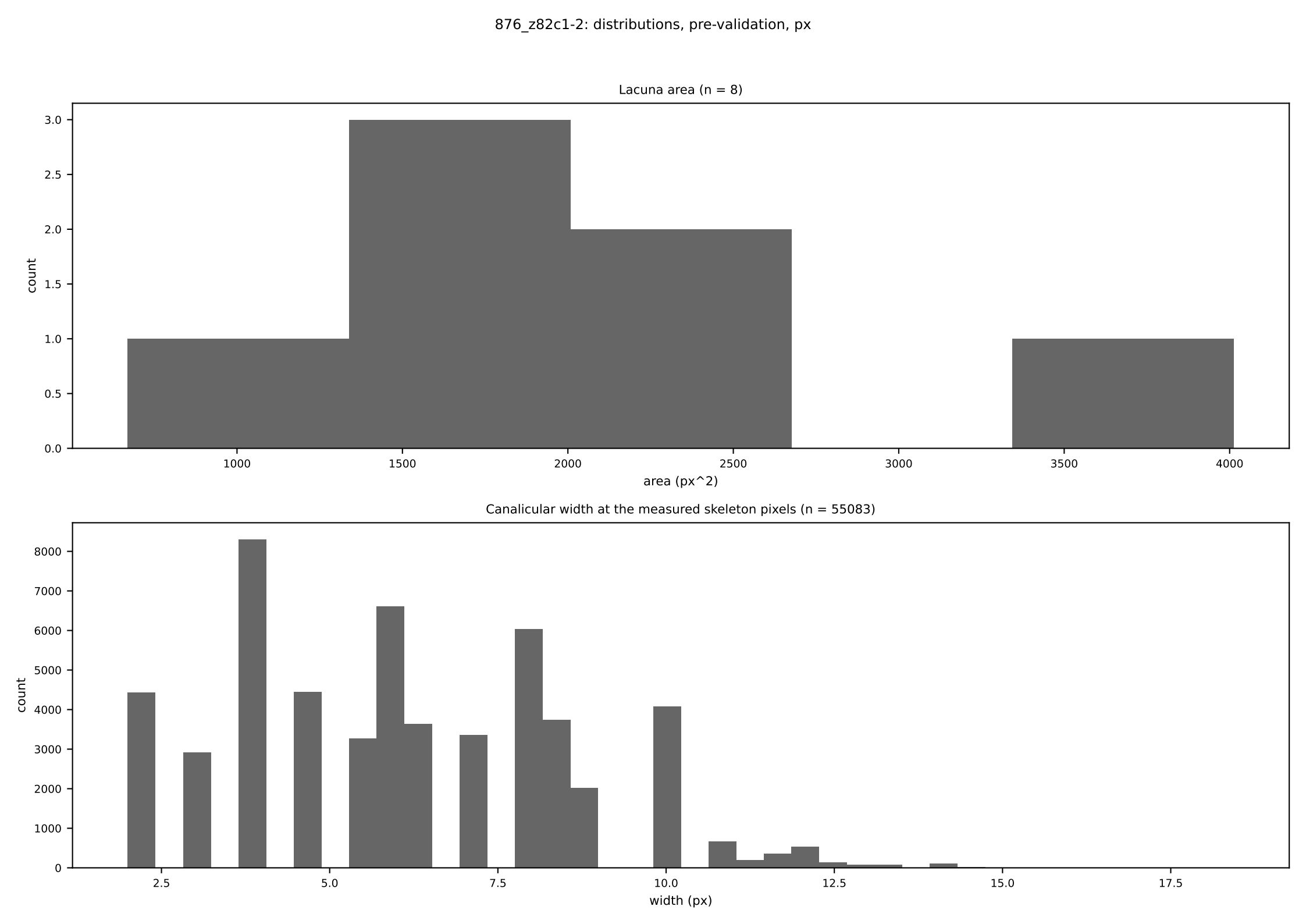
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	26	1091	1899	4505.06	331	13.6105	3
2	15	735	2029	10548.8	727	14.51	7.6575
3	12	757	1961	8241.08	618	13.3351	6.7465
4	11	637	1818	10037.6	756	13.2772	6.8482
5	5	631	1820	6709.48	486	13.8055	6.8296
6	8	505	1245	4383.16	318	13.7835	5.4964
7	6	423	1096	2117.05	165	12.8306	5.9698
8	25	1149	2203	8660.43	622	13.9235	7.1303

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	782.839	12986	23176	0.6542	0.5574	0.0840136	0.0819382	3.3212
2	318.718	10329	26001	1	1	0.0711589	0.0780355	4.7064
3	259.262	9043	23619	1	1	0.0837112	0.0830264	4.6285
4	218.836	8099	21764	1	1	0.0786517	0.0835324	5.0266
5	196.936	7897	21388	1	1	0.0799038	0.0850945	2.5389
6	261.505	9001	23515	1	1	0.0561049	0.0529449	3.0592
7	143.924	6209	15482	0.9898	0.854	0.0681269	0.0707919	4.1689
8	919.704	16560	31079	0.897	0.8325	0.0693841	0.0708839	2.7183

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	1054	1819	1221.34	2116.38	24	26	17
2	698	1770	820.056	2282.39	19	22	23
3	681	1837	850.843	2202.63	13	24	26
4	605	1681	723.973	2059.42	12	19	21
5	588	1610	719.6	2069.25	17	20	18
6	467	971	566.688	1389.92	16	15	17
7	412	904	483.919	1240.49	10	11	20
8	1053	2012	1268.99	2434	32	23	26


876_z82c1-2: distributions, pre-validation, px
Lacuna area (n = 8)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 55083)
count
width (px)